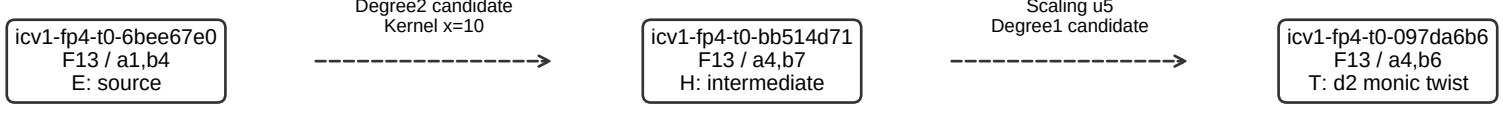PROPOSED: exact models registered; map and subgroup checks pending